M2MVISCODE: MACHINE TO MACHINE VISUAL CODES FOR REPRODUCIBLE GENERATION AND RECOGNITION WITHOUT SEGMENTATION

Jiafan Guo, Kaijin Qiu

Southwest University, Chongqing, China

ABSTRACT

Machine to machine (M2M) visual communication requires reliable recovery of machine generated visual codes from captured images. However, visual code research lacks a controlled and reproducible dataset for stochastic variation, variable image width, and sequence recovery from complete images. To address this problem, we propose **M**achine **to** **M**achine **Vis**ual **Code**s (M2MVisCode), a framework for reproducible generation and recognition, and introduce a new controlled dataset, TPG20k. We generate each sample from a source sequence, a fixed key, a sample seed, and a perturbation seed, producing reproducible visual structures with bounded stochastic variation. Recognizers receive only the complete grayscale image and its valid width; keys, seeds, character boxes, and boundaries are withheld. We study three CTC recognition routes: ConvCTC, M2MVisCode, and TransCTC. M2MVisCode achieves the highest mean ACC across all three datasets. Its relative ACC improvements over the strongest competing model on each dataset are 1.44% on TPG20k, 15.07% on ICDAR2013, and 7.21% on IIIT5K. **Code and data will be released at** https://github.com/jadejeff.

***Index Terms*—** Machine to machine communication, visual code recognition, stochastic visual encoding, sequence recognition

1. INTRODUCTION

Machine to machine visual communication uses images when cameras serve as receivers [1, 2] or information must remain attached to physical objects [3]. An encoder maps a message to an observable pattern, and a decoder recovers the sequence from the resulting image. Existing carriers differ mainly in visual structure and the localization required for decoding.

Scene text recognition has evolved from geometric rectification and recurrent sequence modeling toward architectures using attention and Transformers [4]. MORAN and the TRBA configuration combine geometric rectification with recurrent recognition to normalize irregular text before sequence decoding [5, 6]. SAR and DAN improve visual alignment through spatial or decoupled attention [7, 8], while NRTR and SATRN model long range dependencies with self attention [9, 10]. More recent methods strengthen visual and contextual representations: ViTSTR and SVTR employ Transformer based visual modeling [11, 12], whereas ABINet and PARSeq introduce bidirectional or permutation language modeling [13, 14]. These developments provide representative recognition paradigms for evaluating machine generated visual codes.

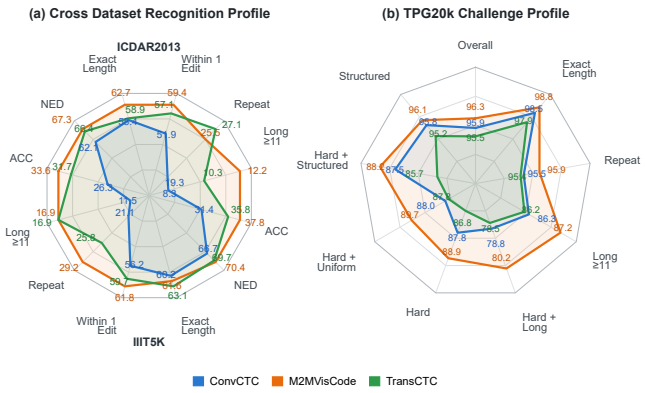


Fig. 1. Recognition profiles of the three M2MVisCode routes across datasets and challenge conditions.

Fig. 1(a) compares the three routes on the public datasets ICDAR2013 and IIIT5K, while Fig. 1(b) shows their profiles under controlled TPG20k challenges. Despite the strong recognition performance reported on widely used scene text datasets such as ICDAR2013 [15] and IIIT5K [16], these methods are designed primarily to recognize naturally occurring or predefined human readable patterns. They do not address how machines can reproducibly generate and recover stochastic visual codes under a fixed key. This limits their direct use in machine to machine visual communication, where visual structures must be controllable, reproducible, and recoverable from complete images. Their original protocols commonly normalize word images to fixed input geometry or employ decoding assumptions developed for natural text. For codes with variable width, repeated symbols, and long sequences, these assumptions may distort content geometry or obscure the valid temporal extent.

To address this problem, we propose M2MVisCode. First, reproducible encoding uses a fixed key for the prototype family; sample and perturbation seeds control bounded prototype variation, ordered composition, and complete

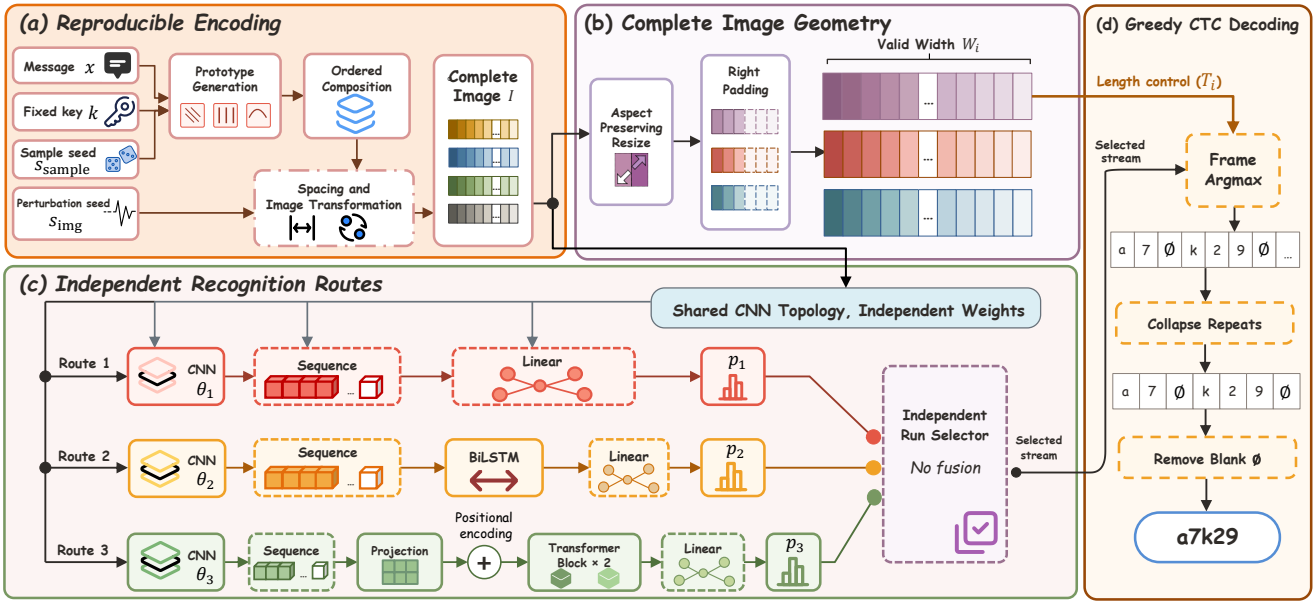


Fig. 2. Overall architecture of M2MVisCode: (a) reproducible encoding; (b) complete image geometry; (c) independent recognition routes; and (d) greedy CTC decoding.

image transformation. Then, each complete image preserves its aspect ratio and is converted into a visual sequence along its width, with unpadded width determining the valid time steps for CTC training and greedy decoding [17]. Three independently trained routes compare convolutional, recurrent, and Transformer modeling without character boxes or boundaries. Finally, we introduce TPG20k, a new dataset with controlled stochastic variation, repeated symbols, long sequences, and structured image degradation.

The main contributions are summarized as follows:

- **Reproducible Code Generation.** We propose a reproducible visual encoding method that uses a fixed key, a sample seed, and a perturbation seed to generate controllable complete visual codes;
- **Recognition without Segmentation.** We develop a complete image recognition method that preserves aspect ratio and maps valid image width to the temporal extent used for sequence training and decoding;
- **A New Controlled Dataset.** We introduce TPG20k for controlled evaluation under stochastic variation, repeated symbols, long sequences, and structured image degradation.

2. METHODOLOGY

A. Overview

Fig. 2 summarizes M2MVisCode. Let $\mathbf{x} = (x_1, \ldots, x_L)$ be a sequence over the alphabet $\mathcal{A}$, which contains 10 digits and 26 lowercase letters. Given K, s_{sample}, and s_{img}, the encoder produces a complete grayscale image I. The recognizer receives only I and its valid width; keys, seeds, character boxes, and boundaries are withheld. The three routes $r \in \{1, 2, 3\}$ share the CNN topology but use separate instances and weights. For controlled architectural comparison, we construct two compact baselines termed ConvCTC and TransCTC. ConvCTC combines the shared convolutional extractor with a classifier applied at each step and the CTC objective [17]. TransCTC introduces Transformer based sequence modeling [4] before CTC classification. M2MVisCode is the proposed method, while a convolutional recurrent architecture is used as its recognition backbone.

B. Reproducible Visual Encoding with Fixed Keys

As shown in Fig. 2(a), a fixed profile maps each key namespace and symbol to a reproducible trajectory family. Local seed s_i depends on the sample seed, position, symbol, and composition condition κ_d for difficulty d:

$$\begin{aligned} s_i &= h(s_{\text{sample}}, i, x_i, \kappa_d), \\ \mathbf{p}_i &= G_\phi(x_i, K; s_i), \end{aligned} \tag{1}$$

Here, (ϕ, K, x_i) selects the prototype family, while s_i adds bounded angle and length variation. Each path begins at the configured center and is extended from accepted endpoints; candidates outside the margin are rejected.

The renderer composes patterns in source order and transforms the complete image:

$$I = \mathcal{R}_{d,s_{\text{img}}}\left(\mathcal{C}_{d,s_{\text{img}}}(\mathbf{p}_1, \ldots, \mathbf{p}_L)\right). \tag{2}$$

In Eq. (2), $\mathcal{C}_{d,s_{\text{img}}}$ composes the prototypes into an ordered code, while $\mathcal{R}_{d,s_{\text{img}}}$ applies image transformations. The difficulty d controls their strength, and s_{img} fixes the stochastic choices, making the complete image I reproducible.

Table 1. Recognition results on TPG20k, ICDAR2013, and IIIT5K. ViTSTR uses width aware readout on the public datasets.

Method	Params (M)	TPG20k (%)			ICDAR2013 (%)			IIIT5K (%)		
		ACC ↑	CER ↓	NED ↑	ACC ↑	CER ↓	NED ↑	ACC ↑	CER ↓	NED ↑
Rosetta [18]	21.327	93.39±0.55	2.15±0.56	98.59±0.35	19.54±7.42	51.19±5.27	47.45±5.11	26.94±1.66	46.95±0.95	54.00±1.29
SATRN [10]	65.560	62.53±6.40	9.40±3.27	92.70±2.44	4.99±1.53	83.67±3.17	22.32±1.64	26.56±1.69	51.29±2.69	53.11±2.17
ABINet [13]	36.739	32.08±7.05	39.34±6.19	70.14±5.47	0.36±0.23	97.17±9.19	9.91±0.44	0.31±0.20	99.04±7.55	11.09±1.89
RobustScanner [19]	47.943	15.72±15.35	39.58±17.50	65.06±16.75	1.12±0.21	137.35±11.71	13.21±0.89	1.53±0.63	139.38±3.31	13.07±1.82
ViTSTR [11]	21.519	11.00±14.24	65.32±27.97	38.65±30.67	0.30±0.10	98.68±7.14	9.66±0.33	0.90±0.15	89.48±1.26	12.03±0.78
ConvCTC [17]	0.203	94.94±0.92	0.67±0.17	99.49±0.12	25.65±3.69	36.60±3.09	61.48±2.73	30.31±1.76	35.04±0.78	66.28±0.80
TransCTC [4]	0.275	94.92±1.23	0.71±0.17	99.48±0.14	29.20±1.94	34.34±0.75	64.68±1.06	35.22±2.46	32.51±1.37	69.42±1.26
M2MVisCode (Ours)	0.287	**96.31±0.63**	**0.48±0.08**	**99.65±0.06**	**33.60±0.84**	**30.83±0.07**	**67.29±0.28**	**37.76±1.96**	**31.15±0.89**	**70.35±1.08**
Δ Improvement		**1.44%**	**28.36%**	**0.16%**	**15.07%**	**10.22%**	**4.04%**	**7.21%**	**4.18%**	**1.34%**

C. Recognition from Complete Images through Valid Width Modeling

As shown in Fig. 2(b), each image is resized to height 32 with preserved aspect ratio, then left aligned and right padded to a batch canvas whose width is a multiple of four. Let W_i be its unpadded width. In Fig. 2(c), each route independently instantiates the same six layer CNN topology with channels $1 \to 16 \to 32 \to 64 \to 64 \to 96 \to 96$; only the first two pooling stages reduce width, for a total factor of four.

For $r \in \{1, 2, 3\}$, CNN extraction and height averaging produce a feature sequence ordered along image width:

$$\begin{aligned} \mathbf{F}_r &= f_{\theta_r}(I) \in \mathbb{R}^{96 \times H' \times T}, \\ \mathbf{z}_{r,t} &= \frac{1}{H'} \sum_{q=1}^{H'} \mathbf{F}_r(:, q, t), \\ \mathbf{Z}_r &= [\mathbf{z}_{r,1}, \ldots, \mathbf{z}_{r,T}] \in \mathbb{R}^{T \times 96}, \end{aligned} \tag{3}$$

where $H' = 2$, T is the padded time dimension, and the parameter sets θ_r are distinct. Unpadded width determines the valid length. For Route 3, $\widetilde{\mathbf{Z}}_3 = \mathrm{PE}(\mathbf{Z}_3 \mathbf{W}_p)$, where $\mathbf{W}_p \in \mathbb{R}^{96 \times 64}$; PE adds sinusoidal positional encoding. The valid lengths and outputs are

$$\begin{aligned} T_i &= \lfloor \lfloor W_i/2 \rfloor /2 \rfloor, \\ \mathbf{H}_r &= \begin{cases} \mathbf{Z}_1, & r = 1, \\ \mathrm{BiLSTM}_{64}(\mathbf{Z}_2), & r = 2, \\ \mathrm{Tr}_{2,4,128}(\widetilde{\mathbf{Z}}_3), & r = 3. \end{cases} \end{aligned} \tag{4}$$

Route 1, ConvCTC, performs direct convolutional classification. Route 2, M2MVisCode, uses one bidirectional LSTM [20] with 64 hidden units in each direction, while Route 3, TransCTC, uses a noncausal Transformer with two layers, four heads, a feedforward dimension of 128, GELU activation, and normalization before each sublayer. Each route has an independent linear classifier. Recognition therefore uses complete images and valid temporal lengths without character boxes, boundaries, or explicit character segmentation.

D. CTC Training and Greedy Decoding

As shown in Fig. 2(d), the classifier produces class probabilities

$$P_r(t, c \mid I) = \mathrm{softmax}(\mathbf{W}_r \mathbf{h}_{r,t} + \mathbf{b}_r)_c, \quad c \in \mathcal{A}_\varnothing. \tag{5}$$

Here, $\mathcal{A}_\varnothing = \mathcal{A} \cup \{\varnothing\}$ contains 36 visible classes indexed 1 to 36 and the blank symbol $\varnothing$ at index zero.

All routes use the CTC objective [17], which marginalizes paths that collapse to $\mathbf{x}$:

$$\mathcal{L}_{\mathrm{CTC}}^{(r)} = -\log \sum_{\boldsymbol{\pi} \in \mathcal{B}^{-1}(\mathbf{x})} \prod_{t=1}^{T_i} P_r(t, \pi_t \mid I). \tag{6}$$

Here, $\boldsymbol{\pi} \in \mathcal{A}_\varnothing^{T_i}$ is a length T_i alignment and $\mathcal{B}$ is the CTC collapse operator.

3. EXPERIMENTS

3.1. Experimental Setup

a) Datasets and Splits. We conduct experiments on two public datasets, ICDAR2013 [15] and IIIT5K [16], and construct TPG20k as a controlled dataset for machine visual codes. TPG20k contains complete machine visual codes with controlled variations in repeated symbols, sequence length, visual difficulty, and structural composition. It contains 15,000 training images, 600 validation images, and 1,200 test images. The supplementary experiments use TPG20k and IIIT5K.

b) Baseline Models. We compare M2MVisCode with seven baselines: ConvCTC [17], TransCTC [4], Rosetta [18], SATRN [10], ABINet [13], RobustScanner [19], and ViTSTR [11].

c) Evaluation Metrics. We use accuracy (ACC), character error rate (CER), and normalized edit distance (NED). Results are reported as the mean ± sample standard deviation (SD) over three independent seeds.

3.2. Quantitative Comparison

Table 1 reports recognition results as the mean and sample SD over three independent seeds. M2MVisCode ranks first

in all nine performance columns. Its largest relative ACC improvement is 15.07% over TransCTC on ICDAR2013. On TPG20k, it reduces CER by 28.36% relative to ConvCTC. On IIIT5K, it improves ACC by 7.21% relative to TransCTC. These results support combining complete image features with bidirectional visual context and CTC alignment for recognition without explicit character segmentation.

3.3. Qualitative Analysis

Fig. 3(a) shows that TPG20k favors a fixed width of 256, while Fig. 3(b) shows that IIIT5K favors 320. These different optima motivate M2MVisCode's geometry preserving input: width follows each image's aspect ratio, preserving symbol proportions and relative spacing, while valid width determines the CTC input length. Thus, recognition accommodates varying image geometries without assuming a universally optimal fixed width.

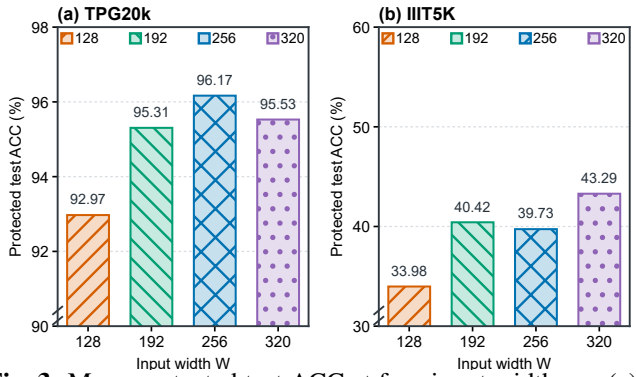


Fig. 3. Mean protected test ACC at four input widths on (a) TPG20k and (b) IIIT5K.

Fig. 4(a) evaluates Gaussian blur over $\sigma = 0$ to 0.5 on TPG20k, while Fig. 4(b) covers $\sigma = 0$ to 1.5 on IIIT5K in an exploratory validation study. M2MVisCode leads both alternatives in mean ACC at every displayed setting, although all routes degrade severely on TPG20k at $\sigma \geq 1$. Its BiLSTM combines preceding and following visual evidence, giving locally blurred symbols additional context for classification. This helps explain its higher recognition accuracy when fine image details are weakened.

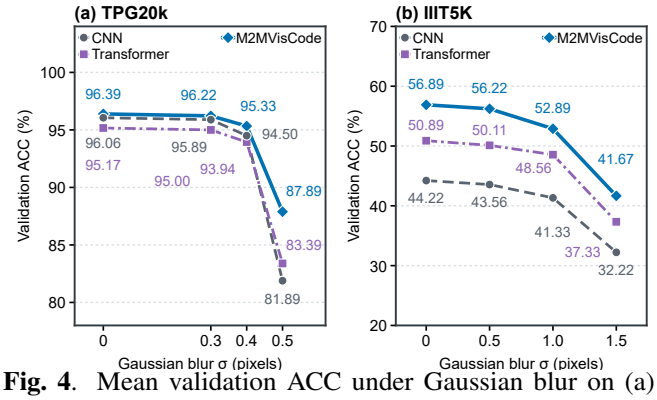


Fig. 4. Mean validation ACC under Gaussian blur on (a) TPG20k and (b) IIIT5K over three seeds.

3.4. Ablation Studies

Table 2 compares three routes with the same CNN topology. M2MVisCode leads all three mean metrics on both datasets, improving IIIT5K ACC by 24.73% relative to CNN while using 0.287M total parameters. These gains cover both complete string recovery and character accuracy. Its BiLSTM integrates local convolutional descriptors with preceding and following visual information before classification, providing contextual support for ambiguous symbols. The results support this combination of local detail and ordered context as an effective design for compact sequence recognition.

Table 2. Sequence route comparison on TPG20k and IIIT5K, reported as mean ± sample SD over three seeds.

Method	Dataset	Params (M)	ACC (%) ↑	CER (%) ↓	NED (%) ↑
CNN	TPG20k	0.203	94.97±0.92	0.67±0.17	99.49±0.12
	IIIT5K		30.28±1.80	35.04±0.78	66.26±0.82
Transformer	TPG20k	0.275	94.92±1.23	0.71±0.17	99.48±0.14
	IIIT5K		35.22±2.45	32.51±1.36	69.42±1.26
M2MVisCode	TPG20k	0.287	**96.31±0.63**	**0.48±0.08**	**99.65±0.06**
	IIIT5K		**37.77±1.97**	**31.15±0.90**	**70.35±1.09**

Fig. 5(a) shows M2MVisCode leading all TPG20k groups, while Fig. 5(b) shows it leading IIIT5K Q2 to Q4, with Transformer leading Q1. Its advantage on longer IIIT5K inputs supports recurrent context aggregation. Useful visual cues may extend beyond a local convolutional neighborhood; the BiLSTM propagates information from both directions to enrich each position's representation. This enables ambiguous local patterns to be interpreted within the broader sequence, helping explain the improved recovery of complete strings from longer visual inputs.

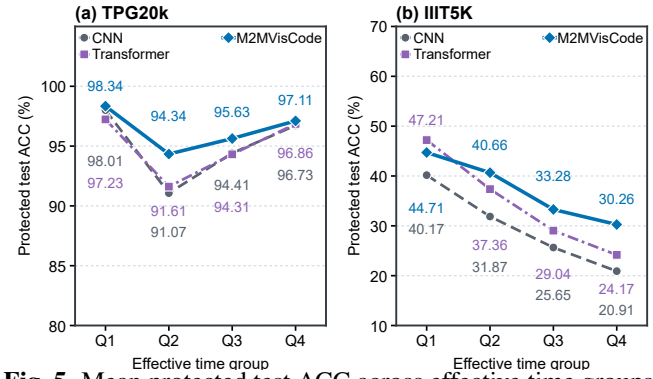


Fig. 5. Mean protected test ACC across effective time groups on (a) TPG20k and (b) IIIT5K.

4. CONCLUSION

This paper presented M2MVisCode, a reproducible framework for generating and recognizing complete machine visual codes without character segmentation. TPG20k enables controlled evaluation across structural and stochastic conditions. Results show reliable recovery on TPG20k and consistent advantages from M2MVisCode across all three datasets. Future work will investigate broader code families and deployment settings.

5. REFERENCES

[1] S. D. Perli, N. Ahmed, and D. Katabi, "PixNet: Interference-free wireless links using LCD-camera pairs," in *Proceedings of the 16th Annual International Conference on Mobile Computing and Networking*, 2010, pp. 137–148.

[2] T. Hao, R. Zhou, and G. Xing, "COBRA: Color barcode streaming for smartphone systems," in *Proceedings of the 10th International Conference on Mobile Systems, Applications, and Services*, 2012, pp. 85–98.

[3] A. Mohan, G. Woo, S. Hiura, Q. Smithwick, and R. Raskar, "Bokode: Imperceptible visual tags for camera-based interaction from a distance," *ACM Transactions on Graphics*, vol. 28, no. 3, Art. 98, pp. 1–8, 2009.

[4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, vol. 30, pp. 5998–6008.

[5] C. Luo, L. Jin, and Z. Sun, "MORAN: A multi-object rectified attention network for scene text recognition," *Pattern Recognition*, vol. 90, pp. 109–118, 2019.

[6] J. Baek, G. Kim, J. Lee, S. Park, D. Han, S. Yun, S. J. Oh, and H. Lee, "What is wrong with scene text recognition model comparisons? dataset and model analysis," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 4715–4723.

[7] H. Li, P. Wang, C. Shen, and G. Zhang, "Show, attend and read: A simple and strong baseline for irregular text recognition," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, pp. 8610–8617, 2019.

[8] T. Wang, Y. Zhu, L. Jin, C. Luo, X. Chen, Y. Wu, Q. Wang, and M. Cai, "Decoupled attention network for text recognition," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 07, pp. 12216–12224, 2020.

[9] F. Sheng, Z. Chen, and B. Xu, "NRTR: A no-recurrence sequence-to-sequence model for scene text recognition," in *2019 International Conference on Document Analysis and Recognition*. IEEE, 2019, pp. 781–786.

[10] J. Lee, S. Park, J. Baek, S. J. Oh, S. Kim, and H. Lee, "On recognizing texts of arbitrary shapes with 2d self-attention," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020, pp. 2326–2335.

[11] R. Atienza, "Vision transformer for fast and efficient scene text recognition," in *Document Analysis and Recognition – ICDAR 2021*, 2021, pp. 319–334.

[12] Y. Du, Z. Chen, C. Jia, X. Yin, T. Zheng, C. Li, Y. Du, and Y.-G. Jiang, "SVTR: Scene text recognition with a single visual model," in *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence*, 2022, pp. 884–890.

[13] S. Fang, H. Xie, Y. Wang, Z. Mao, and Y. Zhang, "Read like humans: Autonomous, bidirectional and iterative language modeling for scene text recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 7098–7107.

[14] D. Bautista and R. Atienza, "Scene text recognition with permuted autoregressive sequence models," in *Computer Vision – ECCV 2022*, 2022, pp. 178–196.

[15] D. Karatzas, F. Shafait, S. Uchida, M. Iwamura, L. Gomez i Bigorda, S. Robles Mestre, J. Mas, D. Fernandez Mota, J. Almazan Almazan, and L. P. de las Heras, "ICDAR 2013 robust reading competition," in *Proceedings of the 12th International Conference on Document Analysis and Recognition*, 2013, pp. 1484–1493.

[16] A. Mishra, K. Alahari, and C. V. Jawahar, "Scene text recognition using higher order language priors," in *Proceedings of the British Machine Vision Conference*, 2012, pp. 127.1–127.11.

[17] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd International Conference on Machine Learning*, 2006, pp. 369–376.

[18] F. Borisyuk, A. Gordo, and V. Sivakumar, "Rosetta: Large scale system for text detection and recognition in images," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018.

[19] X. Yue, Z. Kuang, C. Lin, H. Sun, and W. Zhang, "RobustScanner: Dynamically enhancing positional clues for robust text recognition," in *Computer Vision – ECCV 2020*, 2020, vol. 12364, pp. 135–151.

[20] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional LSTM and other neural network architectures," *Neural Networks*, vol. 18, nos. 5–6, pp. 602–610, 2005.